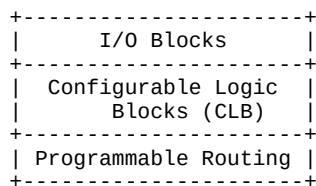


UNIT I - FPGA Based Systems

Introduction to FPGA-Based Systems

FPGA (Field Programmable Gate Array) is a programmable semiconductor device based on configurable logic blocks (CLBs) connected via programmable interconnects. Features: Reconfigurability, parallel processing, high speed, low development cost. Applications: DSP, Embedded systems, Communication systems. Advantages: Short time-to-market, flexibility. Limitations: Higher power consumption compared to ASIC.



FPGA Architectures

Architecture consists of LUTs, Flip-flops, switch matrices, clock management units, BRAM and DSP slices. SRAM-based FPGAs are volatile and widely used. Anti-fuse FPGAs are one-time programmable. Flash-based FPGAs are non-volatile.

